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Perfect synchronization in Sakaguchi–Kuramoto model on directed complex networks

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ABSTRACT

In this study, we analytically derive optimal frequency sets for achieving perfect synchronization at a targeted coupling strength in directed networks of phase-frustrated oscillators. With that aim, we construct a synchrony alignment function (SAF) that helps optimize the synchronization properties of that network and acts as a quantitative indicator of the network's synchronization level. We conduct extensive numerical simulations of the Sakaguchi–Kuramoto (SK) model on both directed scale-free and directed Erdős–Rényi networks to validate the suggested optimal frequency configuration. Numerical simulations demonstrate that the analytically determined frequency set ensures not only stable, perfect synchronization at the desired point in the network but also outperforms other frequency set options in achieving a high degree of synchronization in its vicinity. However, we observe that the synchronization level decreases after reaching perfect synchronization at the targeted point with our derived frequency set. In order to resist this synchronization loss, we give a network reconstruction approach without changing the number of edges in the network. Following the approach, significant improvement in the level of synchronization is achieved. The stability of the system's perfect synchronization state is assessed through a low-dimensional model of the network, while its robustness is evaluated by adding Gaussian noise to the derived frequency set.

2. Analytical derivation

We consider the system of N coupled Sakaguchi–Kuramoto oscillators [25,32] with distributed phase-lags whose dynamics is evolved by

$$\frac{d\theta_i}{dt} = \omega_i + K \sum_{j=1}^N A_{ij} \sin(\theta_j - \theta_i - \alpha_{ij}) \quad (i = 1, 2, \dots, N), \quad (1)$$

where ω_i is node i 's natural frequency and $A = (A_{ij})_{N \times N}$ is the adjacency matrix of a weighted directed network with $A_{ij} = w_{ij}$ if there is an edge from i to j and $A_{ij} = 0$ otherwise, and α_{ij} is the phase lag.

To quantify the level of synchronization in the coupled system, we use the Kuramoto order parameter given by

$$Re^{i\psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}, \quad (2)$$

where $Re^{i\psi}$ denotes the phases' centroid on the complex unit circle, with the magnitude R ranging from 0 (incoherence) to 1 (perfect synchronization).

For the coherent state ($|\theta_j - \theta_i| \rightarrow 0$), using Taylor's expansion we have,

$$\sin(\theta_j - \theta_i - \alpha_{ij}) = \sin(-\alpha_{ij}) + (\theta_j - \theta_i) \cos \alpha_{ij} + \text{h.o.t.},$$

where h.o.t. stands for higher order terms. Then, Eq. (1) can be rewritten using the above Taylor's series expansion and ignoring the higher order terms as,

$$\begin{aligned} \frac{d\theta_i}{dt} = & \omega_i + K \sum_{j=1}^N A_{ij} \sin(-\alpha_{ij}) \\ & - K \left[\sum_{j=1}^N A_{ij} \cos \alpha_{ij} \theta_i - \sum_{j=1}^N A_{ij} \cos \alpha_{ij} \theta_j \right], \end{aligned}$$

which imply

$$\begin{aligned} \frac{d\theta_i}{dt} = & \omega_i + K \sum_{j=1}^N A_{ij} \sin(-\alpha_{ij}) \\ & - K \left[\sum_{j=1}^N \sum_{q=1}^N A_{iq} \cos \alpha_{iq} \delta_{ij} \theta_j - \sum_{j=1}^N A_{ij} \cos \alpha_{ij} \theta_j \right], \end{aligned}$$

i.e.,

$$\frac{d\theta_i}{dt} = \omega_i - K s_i^{\text{out}} - \sum_{j=1}^N L_{ij} \theta_j, \quad (3)$$

where $s_i^{\text{out}} = \sum_{j=1}^N A_{ij} \sin(\alpha_{ij})$ and $L_{ij} = K \sum_{q=1}^N A_{iq} \cos \alpha_{iq} \delta_{ij} - K A_{ij} \cos \alpha_{ij}$.

Now, the Eq. (3) can be written in vector form as,

$$\dot{\theta} = \omega - K s^{\text{out}} - L \theta, \quad (4)$$

where θ , ω , and s^{out} are the column vectors given by $\theta = [\theta_1, \theta_2, \dots, \theta_N]$, $\omega = [\omega_1, \omega_2, \dots, \omega_N]$, and $s^{\text{out}} = [s_1^{\text{out}}, s_2^{\text{out}}, \dots, s_N^{\text{out}}]$ respectively. In steady state, we have $\dot{\theta} = 0$. Therefore, the steady state phases are given by

$$\theta^* = L^\dagger (\omega - K s^{\text{out}}), \quad (5)$$

where $L^\dagger$ is the pseudoinverse of $L = (L_{ij})_{N \times N}$ [47].

Since, L is a real $N \times N$ matrix, then by singular value decomposition (SVD) theorem, there exists orthogonal matrices $U =$

$[u^1, u^2, \dots, u^N]$ and $V = [v^1, v^2, \dots, v^N]$ such that $U^T L V = \Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_N)$ where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_N \geq 0$ [48]. Here, σ_i are the singular values of L and the vectors u^i and v^i are the i^{th} left singular vector and i^{th} right singular vector respectively.

Now, for a strongly connected network, without loss of generality, we can reorder the singular values as $0 = \sigma_1 < \sigma_2 \leq \dots \leq \sigma_N$ and we can rewrite L as $L = \sum_{j=2}^N \sigma_j u^j v^{jT}$. Finally defining $\Sigma^\dagger = \text{diag}(0, \sigma_1^{-1}, \sigma_2^{-1}, \dots, \sigma_N^{-1})$, the pseudoinverse $L^\dagger$ is given by,

$$L^\dagger = V \Sigma^\dagger U^T = \sum_{j=2}^N \frac{v^j u^{jT}}{\sigma_j}. \quad (6)$$

The order parameter in the Eq. (2) is then approximated as

$$R \approx 1 - \frac{1}{2N} \|\theta^*\|^2, \quad (7)$$

where,

$$\begin{aligned} \|\theta^*\|^2 &= \langle L^\dagger(\omega - K s^{\text{out}}), L^\dagger(\omega - K s^{\text{out}}) \rangle \\ &= \sum_{j=2}^N \frac{\langle u^j, \omega - K s^{\text{out}} \rangle^2}{\sigma_j^2}. \end{aligned}$$

Now, substituting the above expression in the Eq. (7), we get,

$$\begin{aligned} R &= 1 - \frac{1}{2N} \sum_{j=2}^N \frac{\langle u^j, \omega - K s^{\text{out}} \rangle^2}{\sigma_j^2} \\ &= 1 - \frac{1}{2} J(\omega, L), \end{aligned} \quad (8)$$

where $J(\omega, L) = \frac{1}{N} \sum_{j=2}^N \frac{\langle u^j, \omega - K s^{\text{out}} \rangle^2}{\sigma_j^2}$ is known as synchrony alignment function (SAF).

Now, from the Eq. (8), it is clear that, $J(\omega, L) \rightarrow 0$ gives $R \rightarrow 1$, which denotes the perfect synchronization state. Therefore, to achieve perfect synchronization at a given point, we choose

$$\omega = K^{(p)} s^{\text{out}}, \quad (9)$$

where $K^{(p)}$ is the targeted coupling strength.

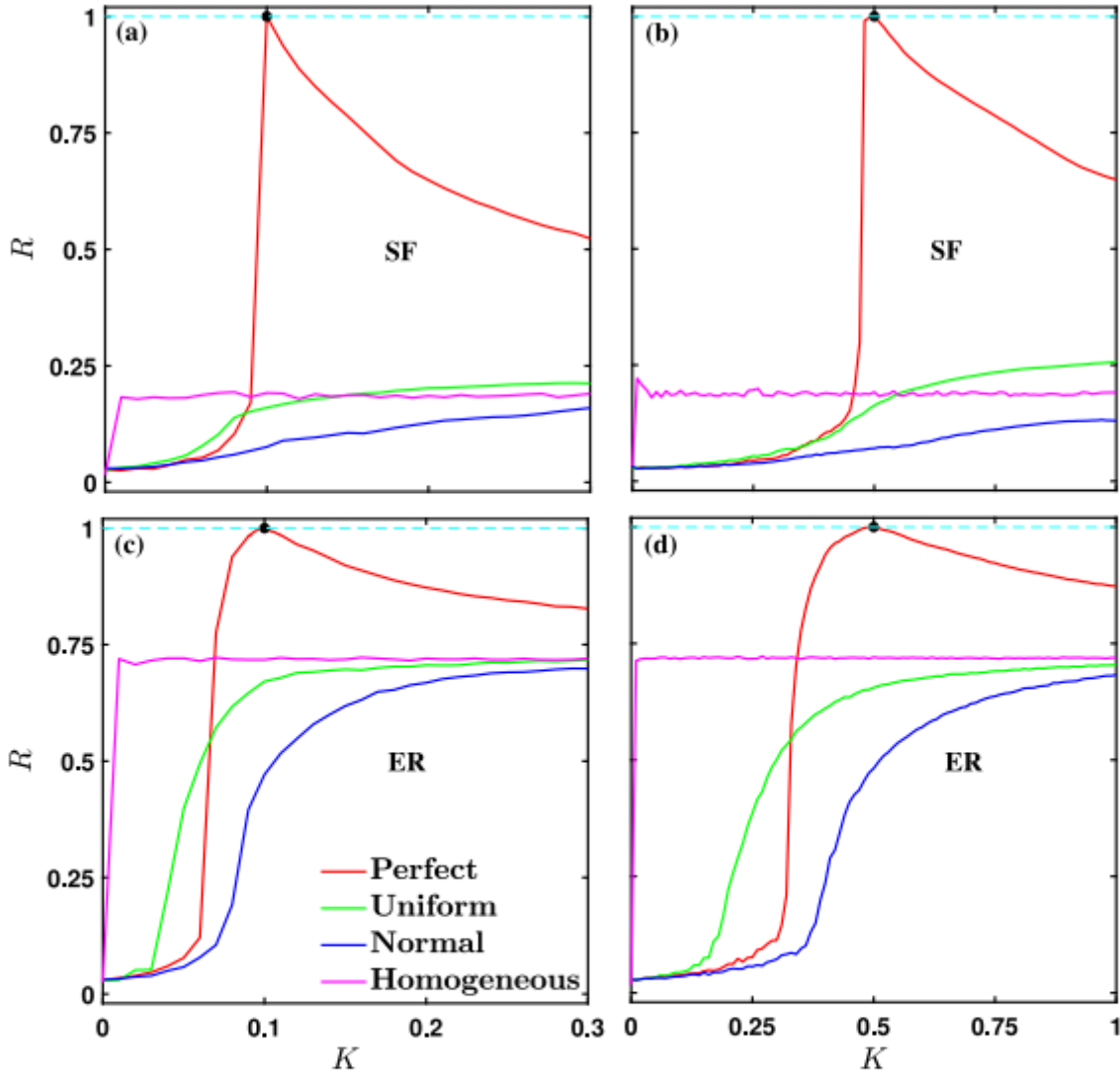


Fig. 2. Synchronization profiles K vs. R for allocation of oscillators at different types of frequency distribution: We have taken weighted SF network and weighted ER network, both of which are of size $N = 1000$, and in both cases, we target at $K^{(p)} = 0.1$ and $K^{(p)} = 0.5$ for perfect synchronization. (a) and (b) represents the synchronization levels for perfect (red), uniform (green), normal (blue), and homogeneous (magenta) for SF Network. In the case of ER network (c) and (d) represent the same as SF case. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)